

EECE 306 Lab 6 notebook. Electric Potential, the Gradient, and Path Independence

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Fill in every TODO, run `publish('Lab06_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

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Setup

```
clear; close all;
e0 = em.const.eps0();
Q = 2e-9;
s = em.src.pointCharge(Q, [0 0 0]);
FE = @(r) em.field.E(s, r);
FV = @(r) em.field.V(s, r);
```

Potential against the closed form

```
radii = (1:10)';
Vnum = em.field.V(s, [radii zeros(10,2)]);
Vref = Q ./ (4*pi*e0*radii);
disp('    r (m)    relative error')
disp([radii abs(Vnum - Vref)./Vref])
```

r (m)	relative error
1.0000	0
2.0000	0
3.0000	0
4.0000	0
5.0000	0
6.0000	0
7.0000	0.0000
8.0000	0
9.0000	0
10.0000	0

E from the gradient, and the error map

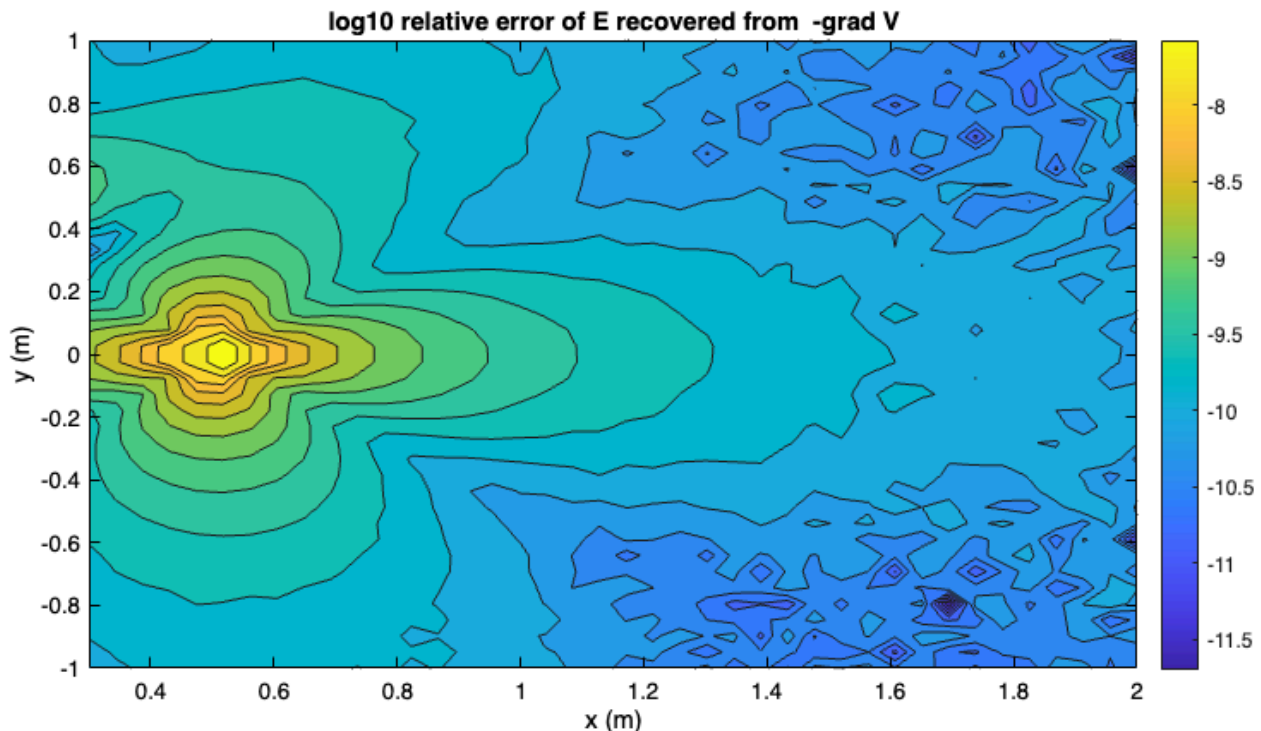
E computed directly and E recovered as minus the gradient of V must agree away from the source.

```
[X, Y] = meshgrid(linspace(0.3, 2, 40), linspace(-1, 1, 40));
P = [X(:) Y(:) zeros(numel(X),1)];
sMap = em.src.merge( ...
```

```

em.src.pointCharge( Q, [-0.5 0 0]), ...
em.src.pointCharge(-Q, [ 0.5 0 0]), ...
em.src.pointCharge(Q/2, [0 0.5 0]));
FVMap = @(r) em.field.V(sMap, r);
Ed = em.field.E(sMap, P);
Eg = -em.op.grad(FVMap, P);
absoluteError = em.vec.mag(Ed - Eg);
fieldMagnitude = em.vec.mag(Ed);
relativeError = absoluteError ./ max(fieldMagnitude, realmin);
errmap = reshape(max(relativeError, realmin), size(X));
figure; contourf(X, Y, log10(errmap), 20); colorbar
xlabel('x (m)'); ylabel('y (m)')
title('log10 relative error of E recovered from -grad V')
% The largest absolute discrepancy is 7.776e-4 V/m near (0.518,-0.026,0) m,
% close to the negative charge where rapid variation increases truncation error.
% The retained relative map shows the same location; the absolute value is
% reported separately because the handout asks for error magnitude.

```



Choosing the step size, the V shaped curve

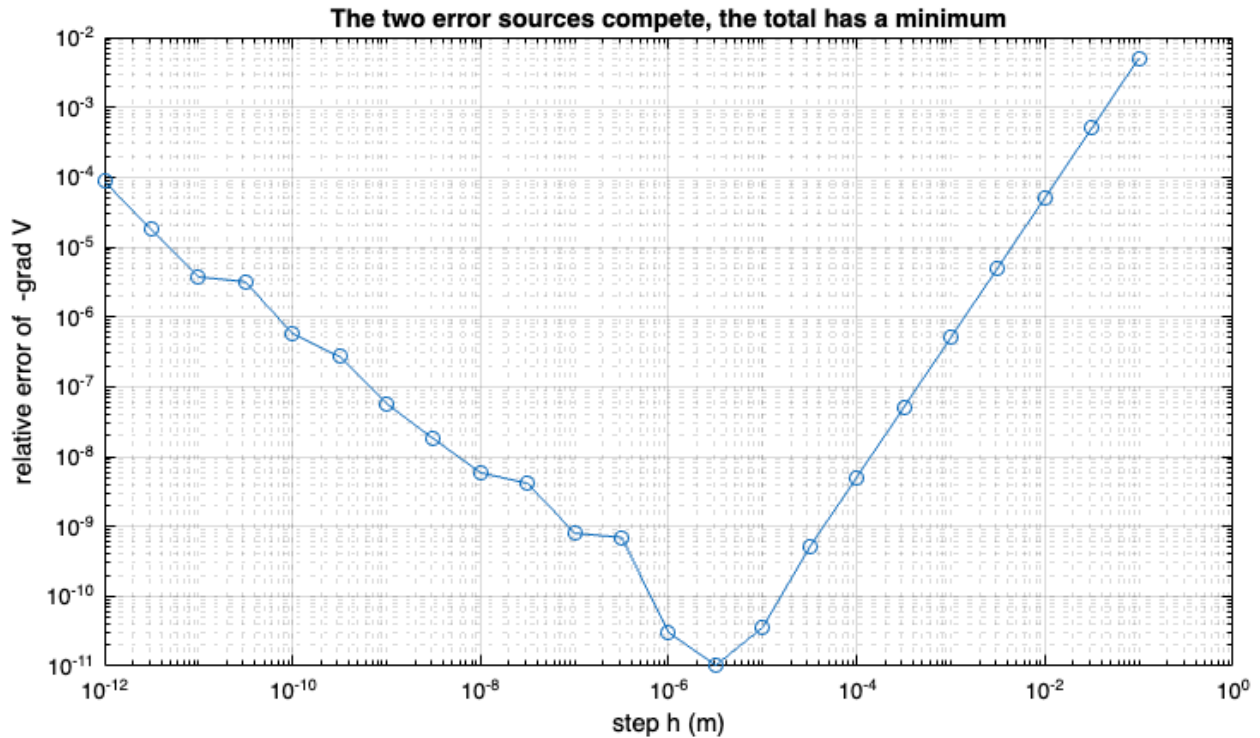
Truncation error falls as h squared, round off grows as 1 over h . The total has a minimum. Sweep h and find it for this machine.

```

p0 = [1 0.5 0];
Eexact = em.field.E(s, p0);
hs = logspace(-12, -1, 23)';
errh = zeros(size(hs));
for k = 1:numel(hs)
    Eh = -em.op.grad(FV, p0, hs(k));
    errh(k) = em.vec.mag(Eh - Eexact) / em.vec.mag(Eexact);
end
figure; loglog(hs, errh, 'o-'); grid on
xlabel('step h (m)'); ylabel('relative error of -grad V')
title('The two error sources compete, the total has a minimum')
[emin, imin] = min(errh);
fprintf('best h on this machine about %.1e with error %.1e\n', hs(imin), emin);
fprintf('the API default h = 1e-5 sits on the flat bottom of this curve\n');

```

best h on this machine about 3.2×10^{-6} with error 1.0×10^{-11}
the API default $h = 1 \times 10^{-5}$ sits on the flat bottom of this curve



The field is conservative, with a control

```
qSquare = @(t) 2*t/pi;
square = @(t) [ ...
    -1+2*min(qSquare(t),1)-2*min(max(qSquare(t)-2,0),1), ...
    -1+2*min(max(qSquare(t)-1,0),1)-2*min(max(qSquare(t)-3,0),1), ...
    zeros(size(t))];
loops = { @(t)[2*cos(t) 2*sin(t) zeros(size(t))], square, ...
    @(t)[1.5+0.5*cos(t) 0.5*sin(3*t) 0.3*sin(t)] };
pathNames = {'circle', 'square', 'nonplanar loop'};
lineC = @(t)[zeros(size(t)) zeros(size(t)) t];
sLine = em.src.lineCharge(1e-9, lineC, [-0.3 0.3], 200);
ball = @(u,v,w)[w*sin(u)*cos(v) w*sin(u)*sin(v) w*cos(u)];
sVol = em.src.volCharge(1e-9, ball, [0 pi], [0 2*pi], [0 0.25], 8, 16, 8);
sources = {sMap, sLine, sVol};
sourceNames = {'point charges', 'finite line', 'volume charge'};
for j = 1:3
    Fsource = @(r) em.field.E(sources{j}, r);
    for k = 1:3
        C = em.field.circulation(Fsource, loops{k}, [0 2*pi], 400);
        fprintf('circulation of E: %s / %s = %.3e V\n', ...
            sourceNames{j}, pathNames{k}, C);
    end
end
Fnc = @(r) [-r(:,2) r(:,1) zeros(size(r,1),1)];
Cnc = em.field.circulation(Fnc, loops{1}, [0 2*pi], 400);
fprintf('circulation of the control field = %.4f (expect 8 pi, clearly nonzero)\n', Cnc);
% The control matters. A test that only ever confirms zero cannot tell
% correct code from code that always returns zero.
```

circulation of E: point charges / circle = -5.272×10^{-12} V
circulation of E: point charges / square = -3.637×10^{-12} V

```

circulation of E: point charges / nonplanar loop = -3.476e-11 V
circulation of E: finite line / circle = 1.880e-12 V
circulation of E: finite line / square = -6.557e-12 V
circulation of E: finite line / nonplanar loop = 4.458e-13 V
circulation of E: volume charge / circle = 2.078e-13 V
circulation of E: volume charge / square = -7.270e-13 V
circulation of E: volume charge / nonplanar loop = 5.830e-14 V
circulation of the control field = 25.1327 (expect 8 pi, clearly nonzero)

```

Potential difference two ways

```

pa = [1 0 0]; pb = [0.3 0.4 0];
dV = em.field.V(s, pb) - em.field.V(s, pa);
seg = @(t) pa + t.*(pb - pa);
work = -em.field.circulation(FE, seg, [0 1], 400);
fprintf('V(b) - V(a) from em.field.V      = %.6e V\n', dV);
fprintf('minus line integral of E, a to b  = %.6e V\n', work);

```

```

V(b) - V(a) from em.field.V      = 1.797510e+01 V
minus line integral of E, a to b  = 1.797513e+01 V

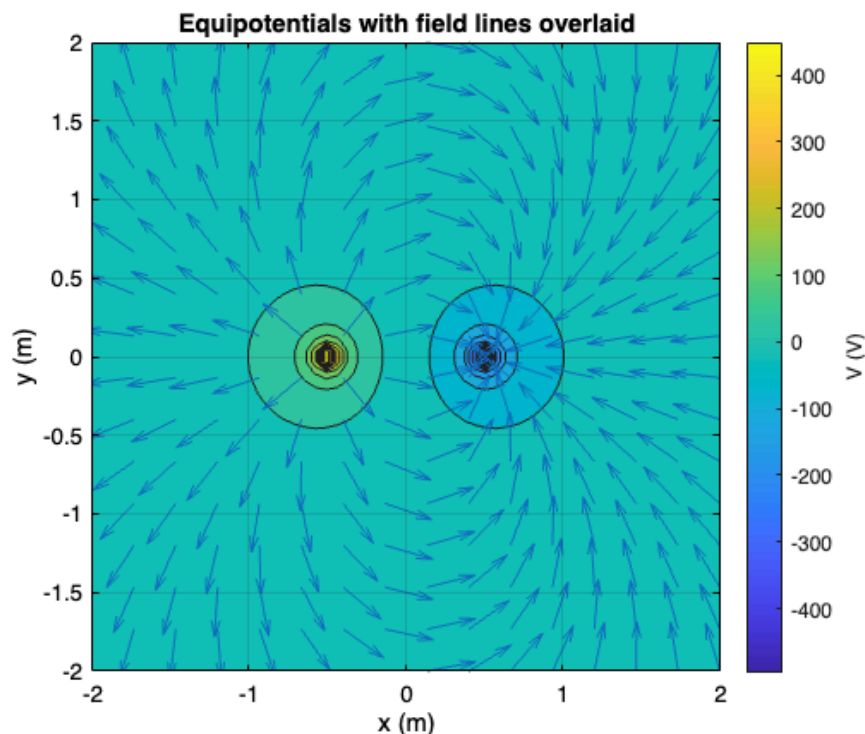
```

Seeing the potential

```

figure;
sDip = em.src.merge( ...
    em.src.pointCharge( Q, [-0.5 0 0]), ...
    em.src.pointCharge(-Q, [ 0.5 0 0]));
em.viz.contourV(@(r) em.field.V(sDip, r), [-2 2], [-2 2], 60, ...
    struct('title', 'Equipotentials of a dipole'));
hold on
em.viz.quiver2(@(r) em.field.E(sDip, r), [-2 2], [-2 2], 16, ...
    struct('normalize', true, 'title', 'Equipotentials with field lines overlaid'));
title('Equipotentials with field lines overlaid')

```



Interpretation

The sampled minimum is $h=3.162\text{e-}6$ m with relative error $1.025\text{e-}11$; $h=1\text{e-}5$ m gives $3.586\text{e-}11$, so it is a useful nearby compromise. Every later gradient should be checked at its own spatial scale because small h eventually amplifies subtraction round-off instead of improving accuracy. All nine closed integrals are below $3.5\text{e-}11$ V, while the control gives 8π . The dipole arrows point toward decreasing potential; a sign error would reverse their direction toward the positive charge while keeping right angles.

Problems encountered

The starter states the default is on a flat bottom, but the sampled minimum is at $3.162\text{e-}6$ m and the default error is 3.50 times larger. The relative map is preserved; realmn guards zero magnitudes and $\log_{10}(0)$, while tests measure the unmodified absolute discrepancy. A finite element list cannot identify infinite-source intent, so V warns when optional logical `s.extentGrowsWithN` is true (set after merging). Fixed-geometry refinement changes V by $4.965\text{e-}6$ V without a warning; doubling line extent changes V from 25.9495 to 37.6527 V. The 400-panel open integral differs by $2.706\text{e-}5$ V; doubling panels quarters this error, and an independent Gauss integral agrees within $1\text{e-}8$ V. Square corners align with midpoint panels; exact-tangent edge sums agree within $4.2\text{e-}11$ V, avoiding an assumption of smoothness at corners.

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
```

```
eta =
```

```
376.7303
```

```
test_lab01 checks complete
```

```
PASS test_lab01          0.10 s
```

```
test_lab02 checks complete
```

```
PASS test_lab02          0.04 s
```

```
5000-point field evaluation: 0.0006 s
```

```
test_lab03 checks complete
```

```
PASS test_lab03          0.04 s
```

```
test_lab04 checks complete
```

```
PASS test_lab04          0.55 s
```

```
merged source: flux 5.00000000e-09 C, enclosed 5.00000000e-09 C
```

```
density errors at 128000 / 1024000 elements: 0.953% / 1.715% max
```

```
sphere / cube / closed hemisphere: 3.00000000e-09 / 3.00000000e-09 / 3.00000000e-09 C
```

```
sphere Gauss rel err 1.42e-11; midpoint N=960 rel err 4.46e-07
```

```
test_lab05 checks complete
```

```
PASS test_lab05          1.80 s
```

```
Lab 6: max absolute map error 7.776e-04 V/m
```

```
max closed residual / (max|E| L) 2.473e-13
```

```
Square vs exact-tangent edges 4.178e-11 V
```

```
open midpoint errors [0.0004329 0.0001082 2.706e-05 6.764e-06] V (order 2)
```

```
Fixed line refinement 4.965e-06 V
```

```
growing line potentials 25.9495 -> 37.6527 V
```

```
warning verified
```

```
Step minimum  $h=3.162\text{e-}6$  m, error  $1.025\text{e-}11$ 
```

```
default error  $3.586\text{e-}11$ 
```

```
10000 points V/grad 0.001/0.003 s
```

```
test_lab06 checks complete
```

```
PASS test_lab06          0.76 s
```

```
6 passed, 0 failed, 6 total
```

```
All tests passing.
```